Final Paper Instructions

Due 12/14

Papers should be about 10 double-spaced pages long and roughly structured as follows:

- **Introduction (~1-2 pages)**
  - Very briefly discuss your research question: why it's important/interesting, maybe some brief literature review and situating the research question in existing evidence, etc. This course does not cover how to ask effective research questions; we only cover how to answer quantitative questions once they have already been posed. Therefore, you won't be evaluated based on how well you motivate your research question. If you want to do an analysis of penguins because they are cute, go for it. But if you're a first year PhD student, this could also be a good opportunity to begin pulling together a bit of literature for a question you might pursue in something like a second-year paper.
- **Data (~2 pages)**
  - Describe your data. You should answer all the following questions in this section:
    - Where did your data come from? Is it publicly available secondary data that you accessed online, or did it involve primary data collection? Describe as much about the data collection process as possible. If data comes from a survey, how was the survey conducted? What was the time period covered? What population was sampled?
    - How did you construct the final variables used in your analysis? How was raw data cleaned, transformed, etc.? Sometimes you just use the raw data if your survey items were already exactly what you need, but sometimes this section can be quite long – for example, if you applied coding procedures to unstructured text data from social media. It should be clearly exactly how you went from raw data to the final variables used in your model.
  - It's typical to include the total sample size in the Data section, but do not include more specific samples sizes of subgroups; this is all something you cover in the Results (e.g., Table 1). Do not include summary statistics in the Data section (e.g., means, medians).
- **Methods (~2 pages)**
  - Describe your specific target statistic that you are trying to infer something about; this must involve at least one additional predictor (or "control") variable. Do this in words, and ideally also using mathematical notation:

    "My target statistic is the average difference in bill depths between Adelie and Chinstrap penguins controlling for bill length:"

$$E[d|s = Adelie, l] - E[d|s = Chinstrap, l]$$

- Describe the regression model that you used to estimate your target statistic.

"I estimated this using a linear regression model taking the following form:"

$$d = \beta_0 + \beta_1(s) + \beta_2(l)$$

- **Results (~2 pages)**
  - Describe your sample and results, referring to tables and figures. You should include at least the following table and figures, but you may include more if you want:
  - Table 1: Describe your data. This should only include basic summary statistics (e.g., means, medians), sample sizes, and a descriptive caption. For example:

Table 1. Descriptive statistics for sample penguins.

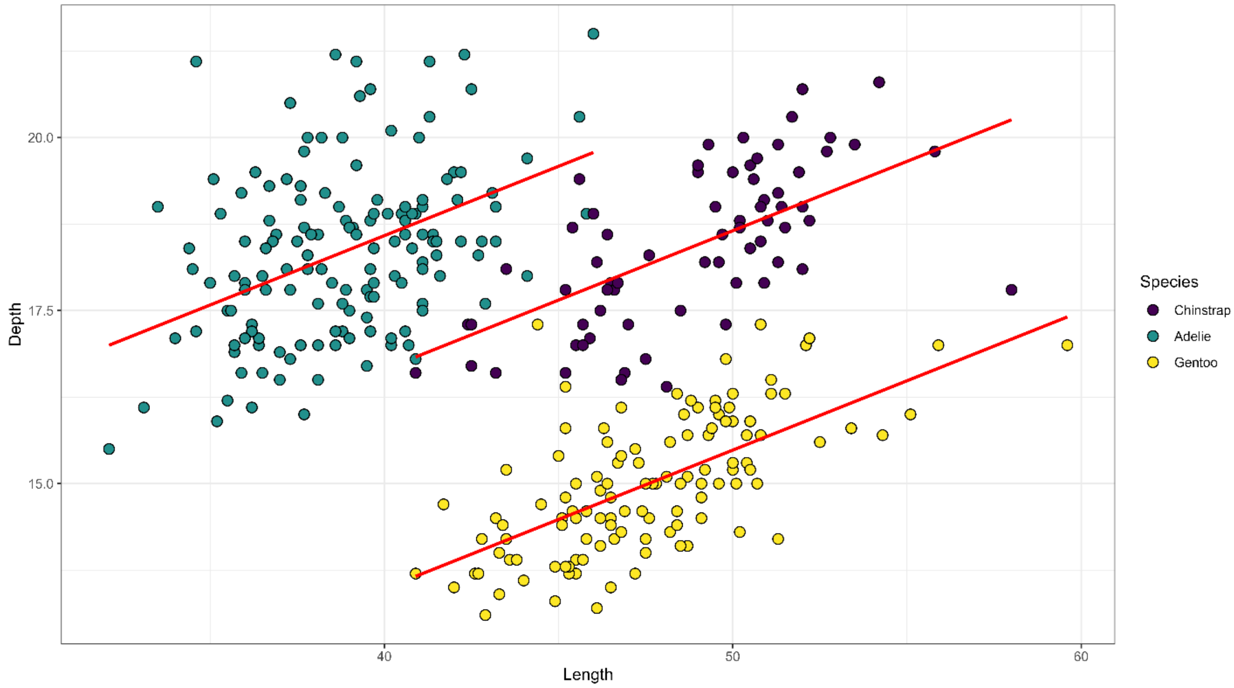
	Adelie	**Chinstrap**	**Gentoo**
Bill depth (mean)	18.3 (1.2)	18.4 (1.1)	15.0 (1.0)
Bill length (mean)	38.8 (2.7)	48.8 (3.3)	47.6 (3.11)
N	**146**	**68**	**119**

  - Table 2: Describe your fitted model. This should include estimates, standard errors (or confidence intervals), and p-values. For example:

Table 2. Parameter estimates from the fitted model predicting penguin bill depth.

	Estimate
Bill length	0.20*** (0.02)
Adelie (reference = Chinstrap)	1.93*** (0.23)
Gentoo (reference = Chinstrap)	-3.17*** (0.15)
Adjusted R^2	**0.77**
N	**333**

- Figure 1: This can be a purely descriptive figure or can visualize model estimates (or both). For example:



- **Discussion (~2 pages)**
  - Put your results in context. This section should cover the following:
    - What have we learned? Summarize your Results section at a high level, usually without using specific statistics.
    - What are the implications, theoretically or empirically? What new studies or data collection might we think about given these results?
    - What are the limitations of your study? Is there perfect data that could overcome these limitations? Are you making certain assumptions that may not be realistic? How do these limitations suggest avenues for future research?
  - One paragraph conclusion: a quick rehash of the overall contribution of this study.